

# Artificial Intelligence (AI)

## Lab Sheet No: 5

### Introduction

Constraint programming is a useful tool in formulating and solving problems that can be defined in terms of constraint among a set of variables. In fact real world problems are defined in terms of some variables that bear some constraints. Finding a set of variables that are within the constraints given (or observed) is a solution to that problem.

Let us consider a problem, that can be represented by some relations of the variables  $x$ ,  $y$  and  $z$ . We have a domain  $D_x$ ,  $D_y$ ,  $D_z$  from where the variables can take a value. The constraint is given by a set  $C$  and may have a number of constraints  $C_1$ ,  $C_2$ ,  $C_3$  etc each relating some or all of the variables  $x$ ,  $y$ ,  $z$ . Now a solution (or solutions ) to the problem is a set of the problem is a set  $dx \in D_x$ ,  $dy \in D_y$ ,  $dz \in D_z$  and all the constraints of set  $C$  are satisfied.

### Crypto arithmetic problem

Crypto Arithmetic Problem is yet another constraint satisfaction problem. We have to assign numeric values (0 through 9) to the alphabets in the given words in such a way that the sum of the two words equals the third.

For example, **SEND+MORE=MONEY**

We have to assign values to the individual alphabets in such a way the arithmetic rules are followed, a trivial solution will be assign zeros to all but we have a constraint, no two alphabets should be assigned with the same number.

|    |    |    |    |   |
|----|----|----|----|---|
| C4 | C3 | C2 | C1 |   |
|    | S  | E  | N  | D |
| +  |    |    |    |   |
|    | M  | O  | R  | E |
|    |    |    |    |   |
| M  | O  | N  | E  | Y |

Now domain for alphabet is given by:

$S, E, N, D, M, O, R, Y \in \{0,1,2,3,4,5,6,7,8,9\}$ .

The constraints are:

$$D+E=Y+10C_1$$

$$N+R+C_1=E+10C_2$$

$$E+O+C2=N+10C3$$

$$S+M+C3=O+10C4$$

$$M=C4$$

$$C1, C2, C3, C4 \in \{0,1\}$$

And we have the constraint that no two alphabets should be assigned to the same number.

### **PROGRAM: 1**

DOMAINS

int\_list=integer\*

PREDICATES

solution(int\_list)

member(integer,int\_list)

CLAUSES

solution([]).

solution([S,E,N,D,M,O,R,Y]):-

    C4=1, %c4 must be 1 otherwise no need to mention M.

    member(C1,[0,1]),

    member(C2,[0,1]),

    member(C3,[0,1]),

    % C1, C2, C3 will have values 0 or 1

    member(E,[0,1,2,3,4,5,6,7,8,9]),

    member(N,[0,1,2,3,4,5,6,7,8,9]),

    member(D,[0,1,2,3,4,5,6,7,8,9]),

    member(M,[0,1,2,3,4,5,6,7,8,9]),

    member(O,[0,1,2,3,4,5,6,7,8,9]),

    member(R,[0,1,2,3,4,5,6,7,8,9]),

    member(Y,[0,1,2,3,4,5,6,7,8,9]),

    member(S,[0,1,2,3,4,5,6,7,8,9]),

    % S,E,N, D, M, O, R, Y will have values between 0 and 9.

    % The values of S,E,N, D, M, O, R, Y must not be equal.

    S<>E, S<>N, S<>D, S<>M, S<>O, S<>R, S<>Y,

    E<>N, E<>D, E<>M, E<>O, E<>R, E<>Y,

    N<>D, N<>M, N<>O, N<>R, N<>Y,

    D<>M, D<>O, D<>R, D<>Y,

    M<>O, M<>R, M<>Y,

    O<>R, O<>Y,

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R<>Y,
% Computation for solution
D+E=Y+10*C1,
N+R+C1=E+10*C2,
E+O+C2=N+10*C3,
S+M+C3=O+10*C4,
M=C4.
member(X, [X|_]). %member(X, [Head|Tail]) is true if X = Head
                  % that is, if X is the head of the list

member(X, [_|Z]):- % or if X is a member of Z,
    member(X,Z).   % ie. if member(X, Z) is true

GOAL
solution([S,E,N,D,M,O,R,Y]).

```

### Assignments

1. Make a program for crypto arithmetic problem as given in Lab and find the solution.
2. Write a program to find a solution for Eight Queens Problem.

### Note: Eight Queens Problem

Eight queens problem is a constraint satisfaction problem. The task is to place eight queens in the 64 available squares in such a way that no queen attacks each other. So the problem can be formulated with variables  $x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8$  and  $y_1, y_2, y_3, y_4, y_5, y_6, y_7, y_8$ ; the  $x$ s represent the rows and  $y$ s the column. Now a solution for this problem is to assign values for  $x$  and  $y$  such that the constraint is satisfied.

The problem can be formulated as  $P = \{(x_1, y_1), (x_2, y_2) \dots (x_8, y_8)\}$   
 where  $(x_1, y_1)$  gives the position of the first queen and so on.

So it can be clearly seen that the domains for  $x_i$  and  $y_i$  are  $D_x = \{1, 2, 3, 4, 5, 6, 7, 8\}$  and  $D_y = \{1, 2, 3, 4, 5, 6, 7, 8\}$  respectively.

The constraints are

- 1) No two queens should be in the same row. That is,  $y_i \neq y_j$  for  $i = 1$  to  $8$ ;  $j = 1$  to  $8$ ;  $i \neq j$
- 2) No two queens should be in the same columns. That is,  $x_i \neq x_j$  for  $i = 1$  to  $8$ ;  $j = 1$  to  $8$ ;  $i \neq j$
- 3) There should not be two queens placed on the same diagonal line.  
 That is,  $(y_i - y_j) \neq \pm (x_i - x_j)$ .

Now a solution to this problem is an instance of P wherein the above mentioned constraints are satisfied.

3. Observe the result of the above program and discuss on the result. Test the goal by placing a few queens explicitly.  
(Try goals like solution  $([c(1,1), c(2,B), c(3,C), c(4,8), c(5,E), c(6,F), c(7,G), c(8,H)]$  etc.)
4. Write a program to find a solution for map coloring problem show below:



Variables  $WA, NT, Q, NSW, V, SA, T$

Domains  $D_i = \{red, green, blue\}$

Constraints: adjacent regions must have different colors

e.g.,  $WA \neq NT$  (if the language allows this), or

$(WA, NT) \in \{(red, green), (red, blue), (green, red), (green, blue), \dots\}$

5. Try to solve the above problem using C, C++, .NET or Java.